Economics 880 - PS5

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Question 1

We solve the household's dynamic programming problem by backward induction, taking prices $(r, w, b) = (0.05, 1.05, 0.20)$ as given. The terminal condition is $V_{N+1}(a, z) = 0$, and agents are restricted to nonnegative asset holdings.

Retirees

For a retired agent of age $j \geqslant J^R$, the Bellman equation is

$$V_j(a) = \max_{a' \geqslant 0} \left\{ u^R\left((1+r)a + b - a'\right) + \beta V_{j+1}(a') \right\},$$

where

$$u^R(c) = \frac{c^{\gamma(1-\sigma)}}{1-\sigma}.$$

We solve this problem numerically over the asset grid and obtain the value function for an agent at model-age 50.

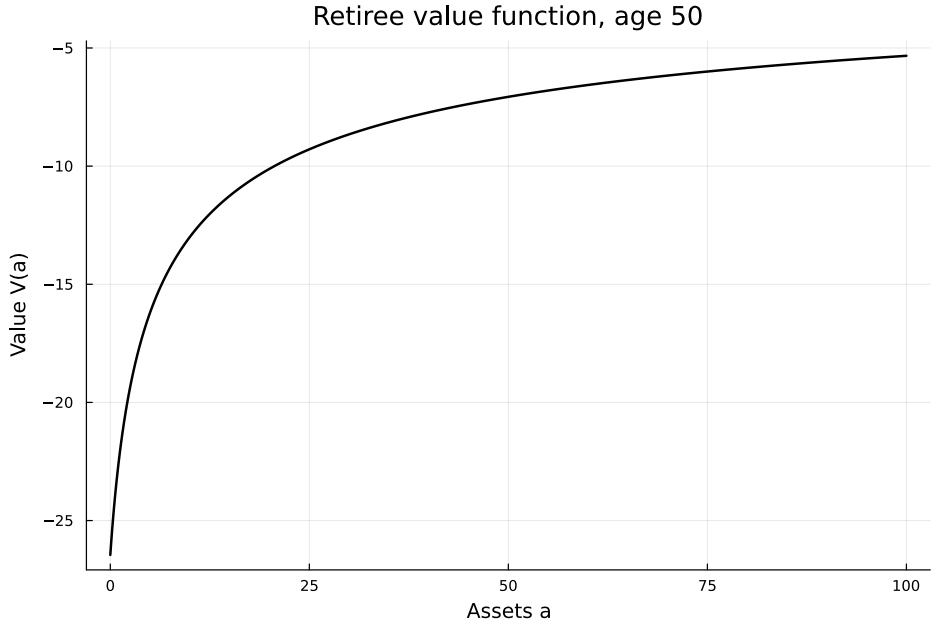


Figure 1: Value function of a retiree at model-age 50.

The value function is increasing in assets, since greater wealth expands the household's feasible consumption set. The underlying continuous-choice problem also implies a concave value function, reflecting diminishing marginal utility of consumption. Our numerical solution is increasing over the entire asset grid and approximately concave.

Workers

For a working-age agent, the Bellman equation is

$$V_j(a, z) = \max_{a' \geqslant 0,\, 0 \leqslant l \leqslant 1} \left\{ u^W(c, l) + \beta \mathbb{E}\left[V_{j+1}(a', z') \mid z \right] \right\},$$

subject to

$$c = (1 + r)a + (1 - \theta) we(z, j) l - a'.$$

To reduce the dimensionality of the optimization, we derive optimal labor supply conditional on the savings choice. The intratemporal first-order condition is

$$\frac{1 - \gamma}{\gamma} \frac{c}{1 - l} = (1 - \theta) we(z, j).$$

Substituting the budget constraint and rearranging yields

$$l_j(a, z, a') = \gamma - \frac{(1 - \gamma)\left[(1 + r)a - a'\right]}{(1 - \theta) we(z, j)},$$

with the solution restricted to $[0, 1]$.

We then solve for the optimal savings policy $a'_j(a, z)$. Figure 2 plots this policy at model-age 20 for both productivity states, $z^H = 3.0$ and $z^L = 0.5$.

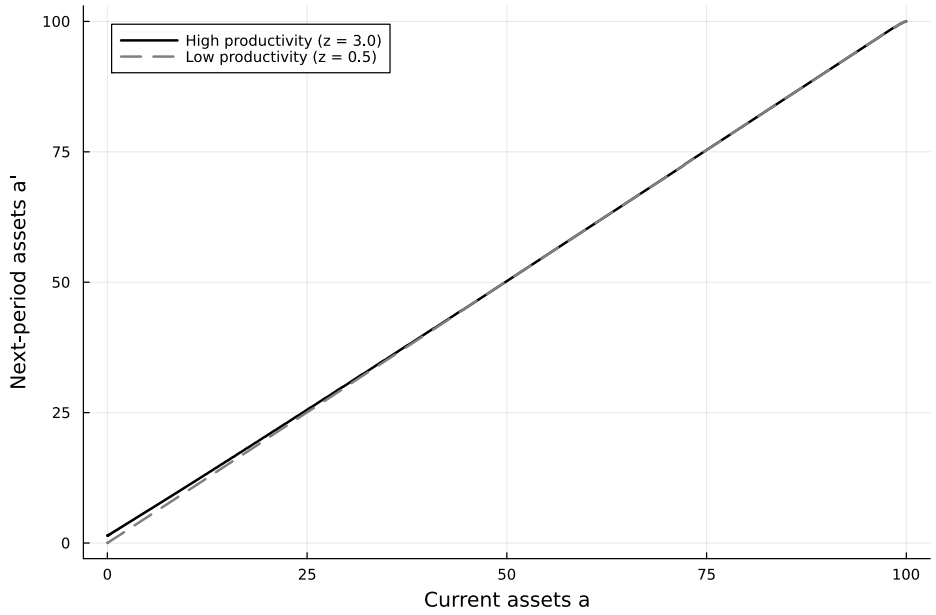


Figure 2: Optimal savings policies at model-age 20 by productivity state.

The numerical results indicate that optimal savings are weakly increasing in current assets for both

productivity states. Moreover, at every point on the asset grid,

$$a'_{20}(a, z^H) \geqslant a'_{20}(a, z^L).$$

Higher-productivity workers earn greater labor income, holding their assets fixed, and consequently choose to save more. Both monotonicity properties are satisfied in our numerical solution.

Question 2

We construct the stationary distribution of agents over age, productivity, and asset holdings using the optimal decision rules obtained in Question 1.

First, the stationary mass of agents at age j satisfies

$$\mu_{j+1} = \frac{\mu_j}{1+n}, \qquad j = 1, \ldots, N-1.$$

Normalizing total population mass to one gives

$$\mu_j = \frac{(1+n)^{-(j-1)}}{\sum_{k=1}^{N} (1+n)^{-(k-1)}}.$$

Newborn agents have zero initial assets, and their productivity is drawn from the stationary distribution of the productivity process. Therefore,

$$F_1(0, z^H) = 0.2037\mu_1,$$
$$F_1(0, z^L) = 0.7963\mu_1.$$

Given the optimal savings policy $a'_j(a, z)$ and productivity transition matrix Π, we recursively compute the distribution at each subsequent age:

$$F_{j+1}(a', z') = \frac{1}{1+n} \sum_{a,z} F_j(a, z)\Pi(z, z')\mathbf{1}\{a'_j(a, z) = a'\}.$$

Here, the indicator function assigns agents to the asset holdings implied by their optimal savings policy, while the transition matrix determines the evolution of productivity.

Our computed distribution satisfies the normalization conditions

$$\sum_{j=1}^{N} \sum_{a,z} F_j(a, z) = 1,$$

and

$$\sum_{a,z} F_j(a,z) = \mu_j \qquad \text{for all } j.$$

Numerically, total population mass is 1.00000000, and the maximum discrepancy between the calculated and theoretical cohort masses is approximately 8.67×10^{-18}.

The newborn distribution is

$$F_1(0, z^H) = 0.00430991,$$
$$F_1(0, z^L) = 0.01684823.$$

Using the stationary distribution, aggregate capital and effective labor supply are calculated as

$$K = \sum_{j=1}^{N} \sum_{a,z} F_j(a,z)a,$$
$$L = \sum_{j=1}^{J^R-1} \sum_{a,z} F_j(a,z)e(z,j)l_j(a,z).$$

At the fixed prices from Question 1, we obtain

$$K = 3.622513, \qquad L = 0.324883.$$

Figure 3 plots average asset holdings by age, conditional on survival to that age.

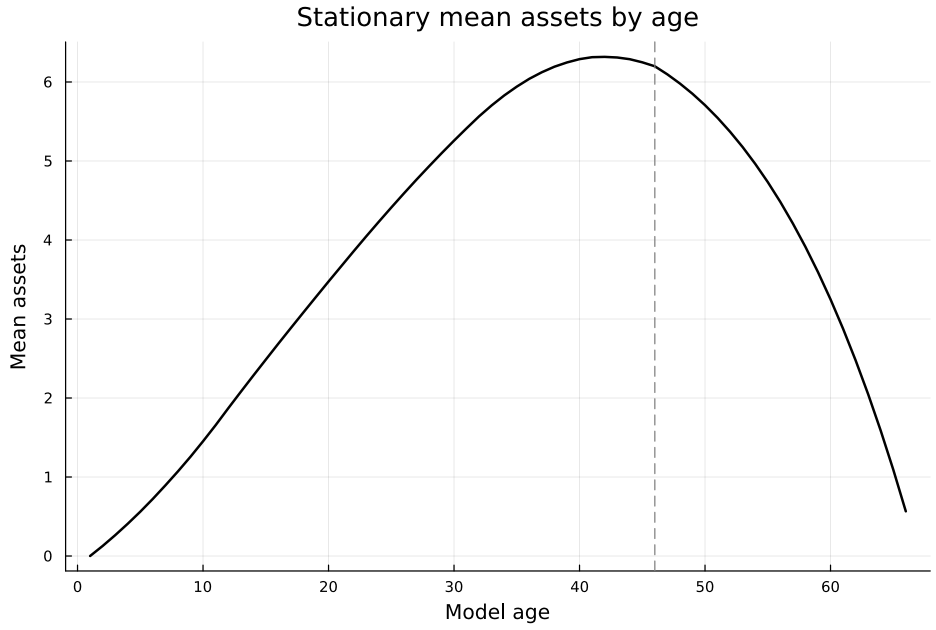


Figure 3: Mean asset holdings by age in the stationary distribution.

Mean asset holdings exhibit a hump-shaped life-cycle profile. Agents accumulate assets during their working years to finance retirement consumption, and subsequently decumulate their wealth after retirement. This pattern reflects the life-cycle savings motive in the model.

Question 3

We solve for stationary general equilibrium under six specifications, using the guess-and-verify algorithm. Table 1 summarizes the results.

Table 1: Steady-state effects of eliminating Social Security

	Benchmark		No risk		Exogenous labor	
	With SS	Without SS	With SS	Without SS	With SS	Without SS
Capital K	3.3584	4.5984	1.0475	1.2975	7.3553	10.4876
Labor L	0.3432	0.3651	0.1601	0.1683	0.7542	0.7543
Wage w	1.4548	1.5931	1.2584	1.3350	1.4530	1.6509
Interest r	0.0236	0.0112	0.0482	0.0374	0.0238	0.0068
Pension benefit b	0.2251	0.0000	0.0909	0.0000	0.4941	0.0000
Total welfare W	-35.7679	-37.4050	-45.0110	-45.1519	-23.0182	-25.7607
cv(wealth)	1.6706	1.4946	0.7953	0.9113	1.6565	1.4177

Benchmark Economy

The benchmark economy is dynamically efficient, since the equilibrium interest rate exceeds population growth, $r = 0.0236 > n = 0.011$. Thus, the implicit return on Social Security is below the return on private capital.

Eliminating Social Security increases aggregate capital by 36.93% and labor supply by 6.39%, raising output by approximately 16.51%. Without pension benefits, households save more to finance retirement, while eliminating the payroll tax increases the incentive to work. The resulting increase in the capital-labor ratio raises wages and lowers the interest rate.

Despite these efficiency gains, aggregate steady-state welfare falls by 1.6370. This suggests that the insurance and redistributive benefits of Social Security outweigh its distortionary costs under our welfare measure. However, newborn welfare increases by 1.6394, with gains for both productivity types. This highlights the distributional consequences of the reform: future generations can benefit from higher capital accumulation, while older generations depend more heavily on pension benefits. The coefficient of variation of wealth also falls from 1.6706 to 1.4946, indicating lower relative wealth inequality.

Economy Without Idiosyncratic Risk

With Social Security, removing idiosyncratic productivity risk reduces aggregate capital from 3.3584 to 1.0475. This is consistent with a precautionary savings motive: when households face uncertain labor income, they accumulate assets as a buffer against adverse productivity shocks. However, the experiment also lowers average productivity by setting both productivity states to 0.5, so the decline in capital cannot be attributed entirely to the removal of risk.

Eliminating Social Security in this economy increases capital by 23.87%, while aggregate welfare falls

by only 0.1410, compared with 1.6370 in the benchmark. This smaller welfare loss is consistent with Social Security providing valuable insurance against idiosyncratic income risk, although the change in mean productivity prevents us from isolating this mechanism.

Importantly, these welfare comparisons concern stationary equilibria rather than the transition following a reform. In particular, they do not account for the losses of existing generations who have paid Social Security taxes but would no longer receive their expected benefits. Consequently, the steady-state welfare differences should not be interpreted as the welfare effects of an immediate reform.

Exogenous Labor Supply

Setting $\gamma = 1$ eliminates the labor-supply distortion associated with Social Security. Consistent with this, aggregate labor is effectively unchanged following the reform, compared with the 6.39% increase in the benchmark.

Social Security nevertheless continues to crowd out private capital accumulation. Eliminating the program increases capital by 42.58%, while aggregate welfare falls by 2.7425.

Thus, even without the labor-supply distortion, Social Security reduces capital accumulation but improves aggregate stationary welfare. The results suggest that the case for Social Security is stronger when labor supply is exogenous, as its insurance and redistributive benefits are obtained without distorting labor supply. Newborn welfare, however, still increases following the reform.